

* Choose the right answer from the given options. [1 Marks Each]

[25]

1. $\cos 35^\circ + \cos 85^\circ + \cos 155^\circ =$

- a. 0
- b. $\frac{1}{\sqrt{3}}$
- c. $\frac{1}{\sqrt{2}}$
- d. $\cos 275^\circ$

Ans. :

- a. 0

Solution:

$$\begin{aligned}
 & \cos 35^\circ + \cos 85^\circ + \cos 155^\circ \\
 &= 2 \cos \left(\frac{35^\circ + 85^\circ}{2} \right) \cos \left(\frac{35^\circ - 85^\circ}{2} \right) + \cos 155^\circ \\
 & \left[\because \cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) \right] \\
 &= 2 \cos 60^\circ \cos(-25^\circ) + \cos 155^\circ \\
 &= 2 \times \frac{1}{2} \cos 25^\circ + \cos 155^\circ \\
 &= \cos 25^\circ + \cos 155^\circ \\
 &= 2 \cos \left(\frac{25^\circ + 155^\circ}{2} \right) \cos \left(\frac{25^\circ - 155^\circ}{2} \right) \\
 &= 2 \cos 90^\circ \cos 65^\circ \\
 &= 0
 \end{aligned}$$

2. If $1, \omega, \omega^2$ are the cube roots of unity then $\omega^2(1+\omega)^3 - (1+\omega^2)\omega =$ [Orissa JEE 2005]

- (A) 1 (B) -1 (C) i (D) 0

Ans. : (D) 0

3. If $\left(\frac{1+i}{1-i} \right)^x = 1$, then:

- a. $x = 2n + 1$
- b. $x = 4n$
- c. $x = 2n$
- d. $x = 4n + 1$, where $n \in \mathbb{N}$

Ans. :

b. $\left(\frac{1+i}{1-i} \right)^x = 1$

Solution:

$$\Rightarrow \left[\frac{(1+i)(1+i)}{(1-i)(1+i)} \right]^x = 1$$

$$\Rightarrow \left[\frac{1+2i+i^2}{1-i^2} \right]^x = 1$$

$$\Rightarrow \left[\frac{2i}{1+1} \right]^x = 1$$

$$\Rightarrow i^x = 1$$

$$\Rightarrow x = 4n, n \in \mathbb{N}$$

4. If z is a complex number, then:

a. $|z^2| > |z|^2$

b. $|z^2| = |z|^2$

c. $|z^2| < |z|^2$

d. $|z^2| \geq |z|^2$

Ans. :

b. $|z|^2 = |z^2|$

Solution:

Let $z = x + yi$

$$|z| = |x + yi| \text{ and } |z|^2 = |x + yi|^2$$

$$\Rightarrow |z|^2 = x^2 + y^2 \dots (i)$$

Now, $z^2 = x^2 + y^2 i^2 + 2xyi$

$$z^2 = x^2 - y^2 + 2xyi$$

$$|z^2| = \sqrt{(x^2 - y^2)^2 + (2xy)^2}$$

$$= \sqrt{x^4 + y^4 - 2x^2y^2 + 4x^2y^2}$$

$$= \sqrt{x^4 + y^4 + 2x^2y^2}$$

$$= \sqrt{(x^2 + y^2)^2}$$

So, $|z|^2 = x^2 + y^2 = |z^2|$

So, $|z|^2 = |z^2|$

5. Let $A = \{(x, y) : y = e^x, x \in \mathbb{R}\}$, $B = \{(x, y) : y = e^{-x}, x \in \mathbb{R}\}$. Then

(A) $A \cap B = \phi$

(B) $A \cap B \neq \phi$

(C) $A \cup B = \mathbb{R}^2$

(D) None of these

Ans. : b

(b) $\because y = e^x, y = e^{-x}$ will meet, when $e^x = e^{-x}$

$$\Rightarrow e^{2x} = 1, \therefore x = 0, y = 1$$

$\therefore A$ and B meet on $(0, 1)$,

$\therefore A \cap B = \phi$.

6. Let $a > 0, a \neq 1$. Then, the set S of all positive real numbers b satisfying

$$(1 + a^2)(1 + b^2) = 4ab \text{ is}$$

- (A) an empty set
- (B) a singleton set
- (C) a finite set containing more than one element
- (D) $(0, \infty)$

Ans. : a

(a)

Given relation

$$(1 + a^2)(1 + b^2) = 4ab$$

$$\Rightarrow a^2 + b^2 - 2ab = 2ab - 1 - a^2b^2$$

$$\Rightarrow (a - b)^2 = -(1 - ab)^2$$

$\because a > 0, a \neq 1$ and b is a positive real number

$\therefore (a - b)^2 \neq 0 \neq -(1 - ab)^2$, because $(a - b)^2$ and $(1 - ab)^2$ are non-negative real numbers

7. Domain of the function $f(x) = \left[\log_{10} \left(\frac{5x - x^2}{4} \right) \right]^{1/2}$ is

- (A) $-\infty < x < \infty$
- (B) $1 \leq x \leq 4$
- (C) $4 \leq x \leq 16$
- (D) $-1 \leq x \leq 1$

Ans. : b

(b) We have $f(x) = \left[\log_{10} \left(\frac{5x - x^2}{4} \right) \right]^{1/2}$ (i)

From (i), clearly $f(x)$ is defined for those values of x for which $\log_{10} \left[\frac{5x - x^2}{4} \right] \geq 0$

$$\Rightarrow \left(\frac{5x - x^2}{4} \right) \geq 10^0 \Rightarrow \left(\frac{5x - x^2}{4} \right) \geq 1$$

$$\Rightarrow x^2 - 5x + 4 \leq 0 \Rightarrow (x - 1)(x - 4) \leq 0$$

Hence domain of the function is $[1, 4]$.

8. If $A = \{1, 2, 4\}, B = \{2, 4, 5\}, C = \{2, 5\}$, then $(A - B) \times (B - C)$ is

- (A) $\{(1, 2), (1, 5), (2, 5)\}$
- (B) $\{(1, 4)\}$
- (C) $(1, 4)$
- (D) None of these

Ans. : b

(b) $A - B = \{1\}$ and $B - C = \{4\}$ $(A - B) \times (B - C) = \{(1, 4)\}$.

9. $\sin 12^\circ \sin 24^\circ \sin 48^\circ \sin 84^\circ =$

- (A) $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$
- (B) $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$
- (C) $\frac{3}{15}$
- (D) None of these

Ans. : a

(a) $\sin 12^\circ \sin 24^\circ \sin 48^\circ \sin 84^\circ$

$$= \frac{1}{4} (2 \sin 12^\circ \sin 48^\circ) (2 \sin 24^\circ \sin 84^\circ)$$

$$= \frac{1}{2} (\cos 36^\circ - \cos 60^\circ) (\cos 60^\circ - \cos 108^\circ)$$

$$= \frac{1}{4} \left(\cos 36^\circ - \frac{1}{2} \right) \left(\frac{1}{2} + \sin 18^\circ \right)$$

$$= \frac{1}{4} \left\{ \frac{1}{4}(\sqrt{5} + 1) - \frac{1}{2} \right\} \left\{ \frac{1}{2} + \frac{1}{4}(\sqrt{5} - 1) \right\} = \frac{1}{16}$$

$$\text{and } \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$$

$$= \frac{1}{2} [\cos (60^\circ - 20^\circ) \cos 20^\circ \cos (60^\circ + 20^\circ)]$$

$$= \frac{1}{2} \left[\frac{1}{4} \cos 3(20^\circ) \right] = \frac{1}{8} \cos 60^\circ = \frac{1}{2} \times \frac{1}{8} = \frac{1}{16}.$$

10. The value of $\sin 47^\circ + \sin 61^\circ - \sin 11^\circ - \sin 25^\circ =$

(A) $\sin 36^\circ$

(B) $\cos 36^\circ$

(C) $\sin 7^\circ$

(D) $\cos 7^\circ$

Ans. : d

(d) $\sin 47^\circ + \sin 61^\circ - (\sin 11^\circ + \sin 25^\circ)$

$$= 2\sin 54^\circ \cos 7^\circ - 2\sin 18^\circ \cos 7^\circ$$

$$= 2 \cos 7^\circ (\sin 54^\circ - \sin 18^\circ)$$

$$= 2 \cos 7^\circ \cdot 2 \cos 36^\circ \cdot \sin 18^\circ$$

$$= 4 \cdot \cos 7^\circ \cdot \frac{\sqrt{5}+1}{4} \cdot \frac{\sqrt{5}-1}{4} = \cos 7^\circ.$$

11. If $\sin A, \cos A$ and $\tan A$ are in $G.P.$, then $\cos^3 A + \cos^2 A$ is equal to

(A) 1

(B) 2

(C) 4

(D) None of these

Ans. : a

(a) We have $\sin A, \cos A$ and $\tan A$ are in $G.P.$

$$\cos^2 A = \sin A \tan A = \frac{\sin^2 A}{\cos A}$$

$$\Rightarrow \cos^3 A - \sin^2 A = 0$$

$$\text{Hence } \cos^3 A + \cos^2 A = \sin^2 A + \cos^2 A = 1$$

12. $|x + \frac{1}{x}| \geq 2$ then $x \in$

(A) $\mathbb{R} - \{0\}$

(B) $\mathbb{R} - \{\pm 1\}$

(C) $\mathbb{R}$

(D) 0

Ans.: (A) $\mathbb{R} - \{0\}$

13. The solution set of $|x - 1| + |x - 2| < 3$ is...

(A) (0,3)

(B) (1,2)

(C) (0,2)

(D) (2,3)

Ans.: (A) (0,3)

14. If $P(n, r) = 1680$ and $C(n, r) = 70$, then $69n + r! =$

(A) 128

(B) 576

(C) 256

(D) 625

Ans. : b

(b) $P(n, r) = 1680 \frac{n!}{(n-r)!} = 1680 \dots (i)$

$$C(n, r) = 70 \Rightarrow \frac{n!}{r!(n-r)!} = 70 \dots (ii) \quad \frac{1680}{r!} = 70, [\text{From (i) and (ii)}]$$

$$r! = \frac{1680}{70} = 24 \Rightarrow r = 4$$

$$\therefore p(n, 4) = 1680$$

$$\therefore n(n-1)(n-2)(n-3) = 1680 \Rightarrow n = 8$$

$$8 \times 7 \times 6 \times 5 = 1680$$

$$\text{Now } 69n + r! = 69 \times 8 + 4!$$

$$= 552 + 24 = 576.$$

15. In how many ways can 6 persons be selected from 4 officers and 8 constables, if at least one officer is to be included
 (A) 224 (B) 672 (C) 896 (D) None of these

Ans. : c

(c) Required number of ways

$$= {}^4C_1 \times {}^8C_5 + {}^4C_2 \times {}^8C_4 + {}^4C_3 \times {}^8C_3 + {}^4C_4 \times {}^8C_2 \\ = 4 \times 56 + 6 \times 70 + 4 \times 56 + 1 \times 28 = 896 .$$

16. A dictionary is printed consisting of 7 lettered words only that can be made with a letter of the word *CRICKET*. If the words are printed at the alphabetical order, as in an ordinary dictionary, then the number of word before the word *CRICKET* is
 (A) 530 (B) 480 (C) 531 (D) 481

Ans. : a

(a) The number of words before the word *CRICKET* is $4 \times 5! + 2 \times 4! + 2! = 530$.

17. How many numbers can be formed from the digits 1, 2, 3, 4 when the repetition is not allowed
 (A) 4P_4 (B) 4P_3
 (C) ${}^4P_1 + {}^4P_2 + {}^4P_3$
 (D) ${}^4P_1 + {}^4P_2 + {}^4P_3 + {}^4P_4$

Ans. : d

(d) Number of 1 digit numbers $= {}^4P_1$

Number of 2 digit numbers $= {}^4P_2$

Number of 3 digit numbers $= {}^4P_3$

Number of 4 digit numbers $= {}^4P_4$

Hence the required number of ways

$$= {}^4P_1 + {}^4P_2 + {}^4P_3 + {}^4P_4.$$

18. The term independent of y in the expansion of $(y^{-1/6} - y^{1/3})^9$ is
 (A) 84 (B) 8.4 (C) 0.84 (D) -84

Ans. : d

$$(d) (9 - r) \left(-\frac{1}{6}\right) + r \left(\frac{1}{3}\right) = 0 \Rightarrow r = 3$$

So the term independent of y

$$= {}^9C_3 (y^{-1/6})^6 (-y^{1/3})^3 = -84.$$

19. Coefficient of x in the expansion of $\left(x^2 + \frac{a}{x}\right)^5$ is

(A) $9a^2$

(B) $10a^3$

(C) $10a^2$

(D) $10a$

Ans. : b

(b) In the expansion of $(x^2 + \frac{a}{x})^5$ the general term is $T_{r+1} = {}^5C_r (x^2)^{5-r} (\frac{a}{x})^r$
 $= {}^5C_r a^r x^{10-3r}$

Here, exponent of x is $10 - 3r = 1$

$$\Rightarrow r = 3. \therefore T_{2+1} = {}^5C_3 a^3 x = 10a^3 \cdot x$$

Hence coefficient of x is $10a^3$.

20. 6^{th} term in expansion of $(2x^2 - \frac{1}{3x^2})^{10}$ is

(A) $\frac{4580}{17}$

(B) $-\frac{896}{27}$

(C) $\frac{5580}{17}$

(D) None of these

Ans. : b

(b) Applying $T_{r+1} = {}^nC_r x^{n-r} a^r$ for $(x+a)^n$

$$\text{Hence } T_6 = {}^{10}C_5 (2x^2)^5 \left(-\frac{1}{3x^2}\right)^5$$

$$= -\frac{10!}{5!5!} 32 \times \frac{1}{243} = -\frac{896}{27}$$

21. The number of non-zero terms in the expansion of $(1 + 3\sqrt{2}x)^9 + (1 - 3\sqrt{2}x)^9$ is

(A) 9

(B) 0

(C) 5

(D) 10

Ans. : c

(c) Given expression

$$= 2 [1 + {}^9C_2 (3\sqrt{2}x)^2 + {}^9C_4 (3\sqrt{2}x)^4 + {}^9C_6 (3\sqrt{2}x)^6 + {}^9C_8 (3\sqrt{2}x)^8]$$

The number of non-zero terms is 5.

22. If a^2, b^2, c^2 be in A.P., then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ will be in

(A) A.P.

(B) G.P.

(C) H.P.

(D) None of these

Ans. : a

(a) Since a^2, b^2, c^2 be in A.P.

$$\text{Then } b^2 - a^2 = c^2 - b^2$$

$$\Rightarrow (b-a)(b+a) = (c-b)(c+b)$$

$$\Rightarrow \frac{b-a}{c+b} = \frac{c-b}{b+a}$$

$$\Rightarrow \frac{(b-a)(a+b+c)}{(c+a)(b+c)} = \frac{(c-b)(a+b+c)}{(a+b)(c+a)}$$

$$\Rightarrow \frac{b^2+bc-ac-a^2}{(c+a)(b+c)} = \frac{c^2+ac-ab-b^2}{(a+b)(c+a)}$$

$$\Rightarrow \frac{b}{c+a} - \frac{a}{b+c} = \frac{c}{a+b} - \frac{b}{c+a}$$

Hence $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ be in A.P.

23. If the roots of the equation $x^3 - 12x^2 + 39x - 28 = 0$ are in A.P., then their common difference will be

(A) ± 1

(B) ± 2

(C) ± 3

(D) ± 4

Ans. : c

(c) Let $a - d, a, a + d$ be the roots of the equation $x^3 - 12x^2 + 39x - 28 = 0$

Then $(a - d) + a + (a + d) = 12$ and $(a - d)a(a + d) = 28$

$$\Rightarrow 3a = 12 \text{ and } a(a^2 - d^2) = 28$$

$$\Rightarrow a = 4 \text{ and } a(a^2 - d^2) = 28$$

$$\Rightarrow 16 - d^2 = 7$$

$$\Rightarrow d = \pm 3.$$

24. Consider a sequence whose sum of first n -terms is given by

$S_n = 4n^2 + 6n, n \in N$, then T_{15} of this sequence is -

(A) 118

(B) 120

(C) 122

(D) 86

Ans. : c

$$T_n = S_n - S_{n-1}$$

$$\Rightarrow T_n = (4n^2 + 6n) - (4(n-1)^2 + 6(n-1))$$

$$T_n = 8n + 2 \Rightarrow T_{15} = 122$$

25. Maximum value of sum of arithmetic progression 50, 48, 46, 44, is :-

(A) 325

(B) 648

(C) 652

(D) 650

Ans. : d

For maximum sum $\Rightarrow T_n = 0$

$$a + (n-1)d = 0$$

$$\Rightarrow 50 + (n-1)(-2) = 0 \Rightarrow n = 26$$

$$\text{So } S_{26} = \frac{26}{2}[2 \times 50 + 25 \times (-2)] = 650$$
